

The Impact of Patient Clinical Information and Color Constancy Preprocessing on Automated Skin Cancer Detection Using a Fusion Deep Learning Model

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PAD-UFES-20 Dataset • EfficientNetB3 + Squeeze-and-Excitation Clinical Fusion • Patient-Level Stratified 5-Fold Cross-Validation

Introduction & Motivation

- Skin cancer accounts for roughly one in three cancer diagnoses worldwide (WHO), with a similar share of the total cancer burden reported in Brazil.
- Dermatoscopy accuracy depends heavily on examiner experience, and many regions face a shortage of both dermatologists and dermatoscopes.
- Goal: a computer-aided diagnosis (CAD) system that works from ordinary smartphone photographs, fusing clinical data with image features.
- This work extends the PAD-UFES-20 baseline (Pacheco & Krohling) with (1) gray-world color constancy, (2) an attention-based fusion architecture, and (3) a leakage-free, patient-level validation protocol.

Dataset

2,298 clinical images from 1,373 patients (1,641 unique lesions) across six diagnostic categories, collected via smartphone by the PAD program at UFES, Brazil.

Diagnostic category	N of images
Basal Cell Carcinoma (BCC)	845
Actinic Keratosis (ACK)	730
Nevus (NEV)	244
Seborrheic Keratosis (SEK)	235
Squamous Cell Carcinoma (SCC)	192
Melanoma (MEL)	52
Total	2,298

Table 1. Number of samples per diagnostic category in the dataset used in this study.

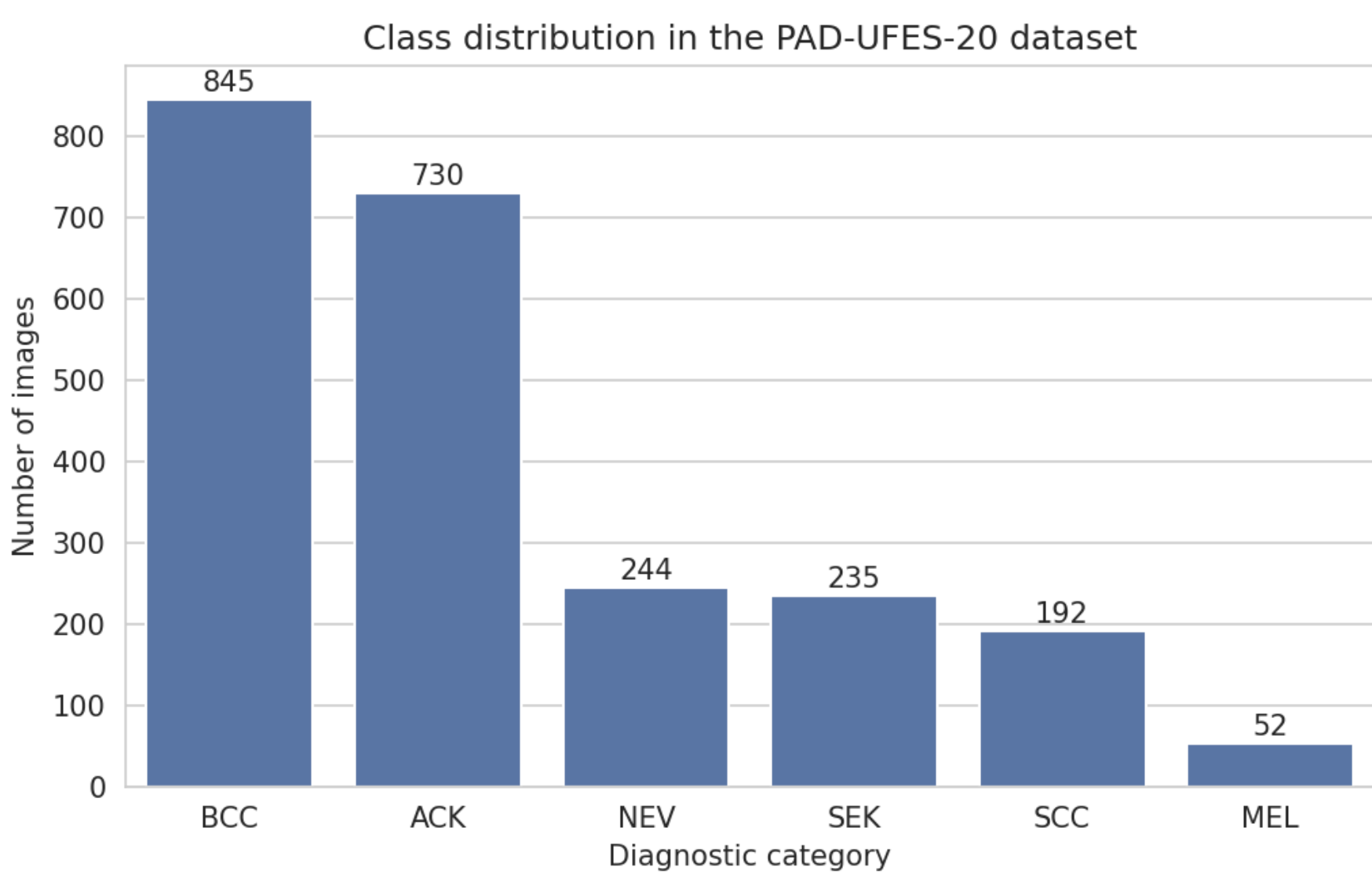


Fig. 1. Class distribution across the six diagnostic categories.

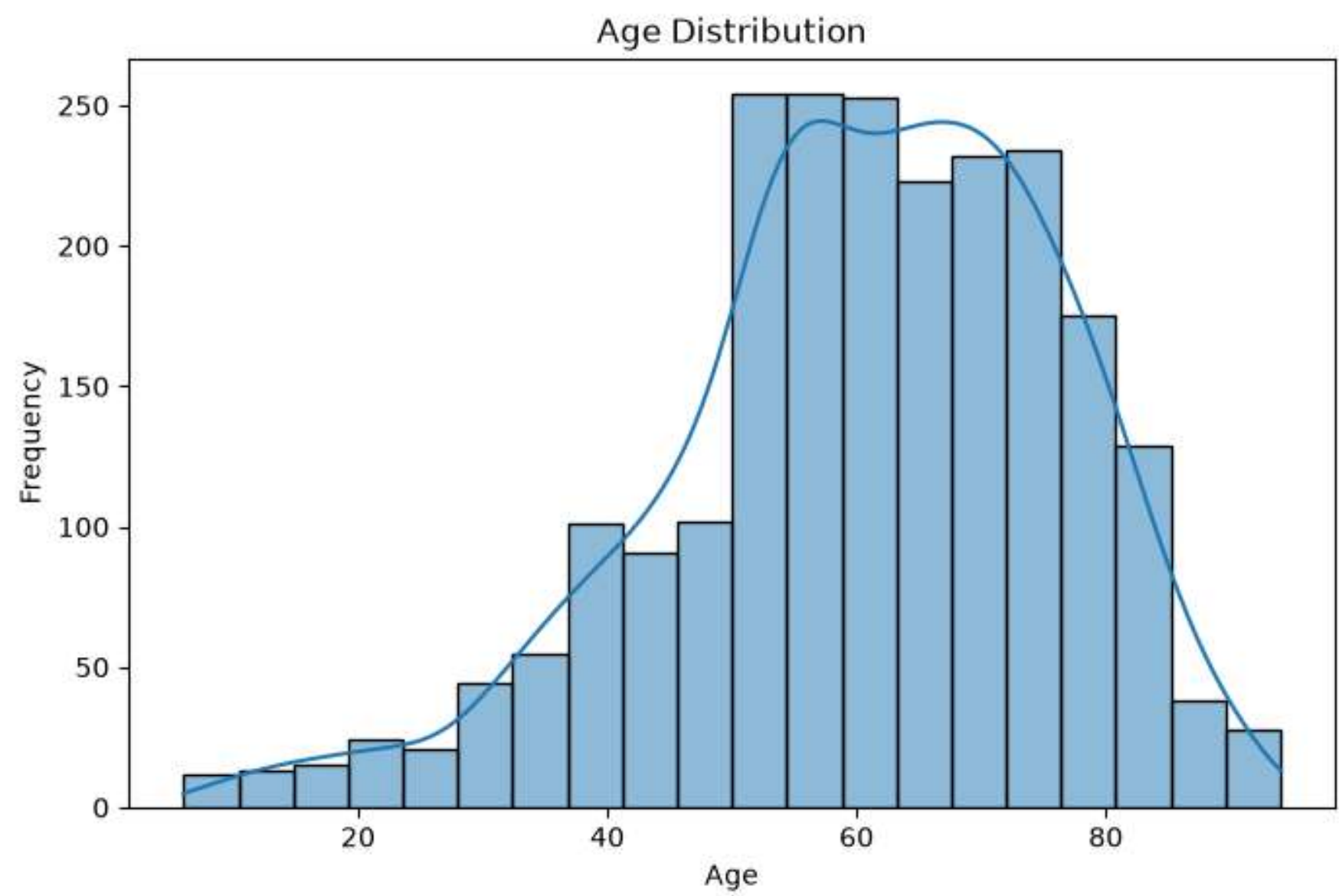


Fig. 2. Age distribution of patients in the dataset.



Fig. 3. Representative preprocessed samples for each diagnostic category.

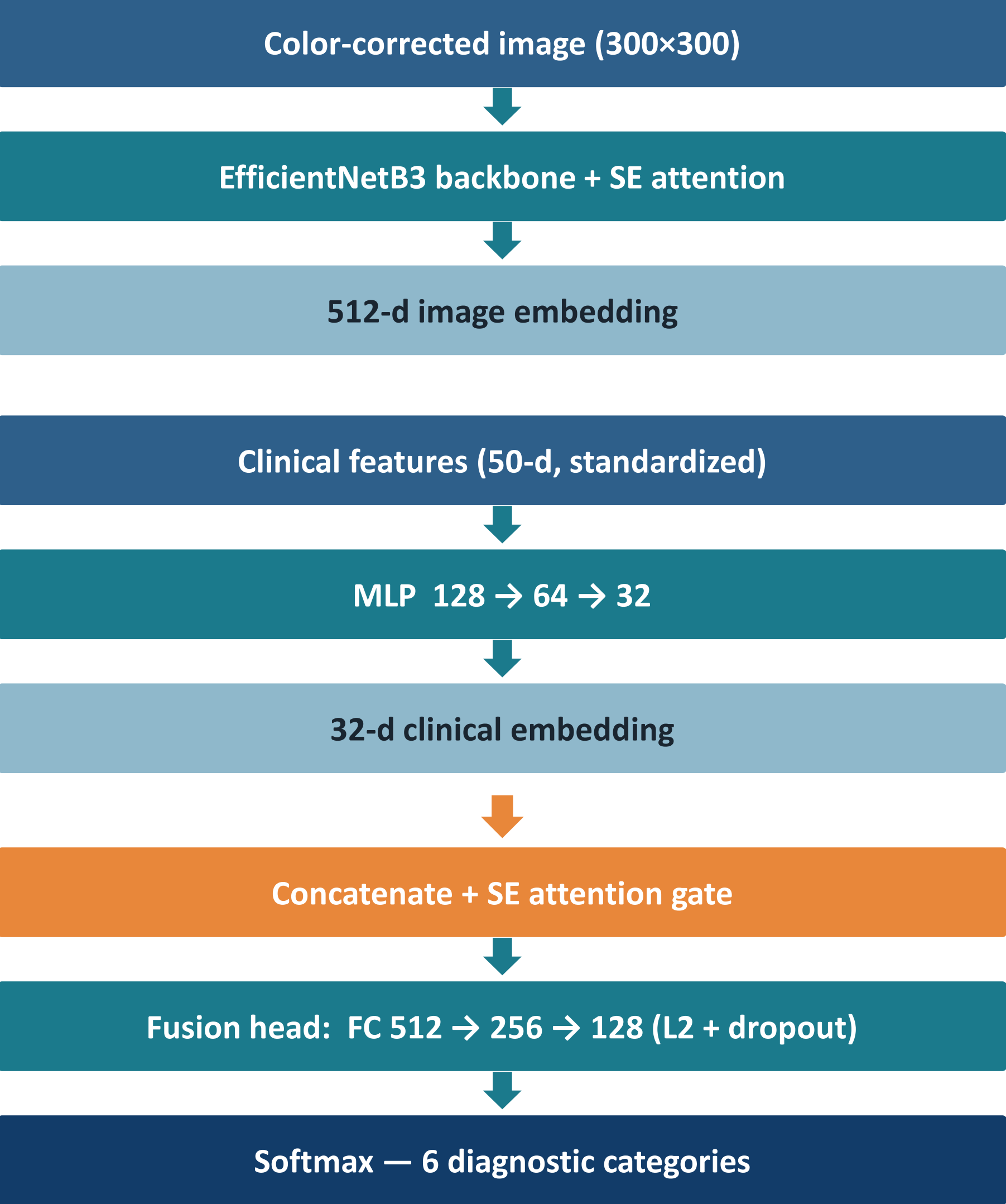
Color Constancy Preprocessing

A gray-world algorithm rescales each color channel so its mean matches the image's overall gray level, correcting for lighting and camera differences across the heterogeneous smartphone images before feature extraction.



Fig. 4. Effect of gray-world color constancy: original (left) versus color-corrected (right).

Proposed Fusion Architecture

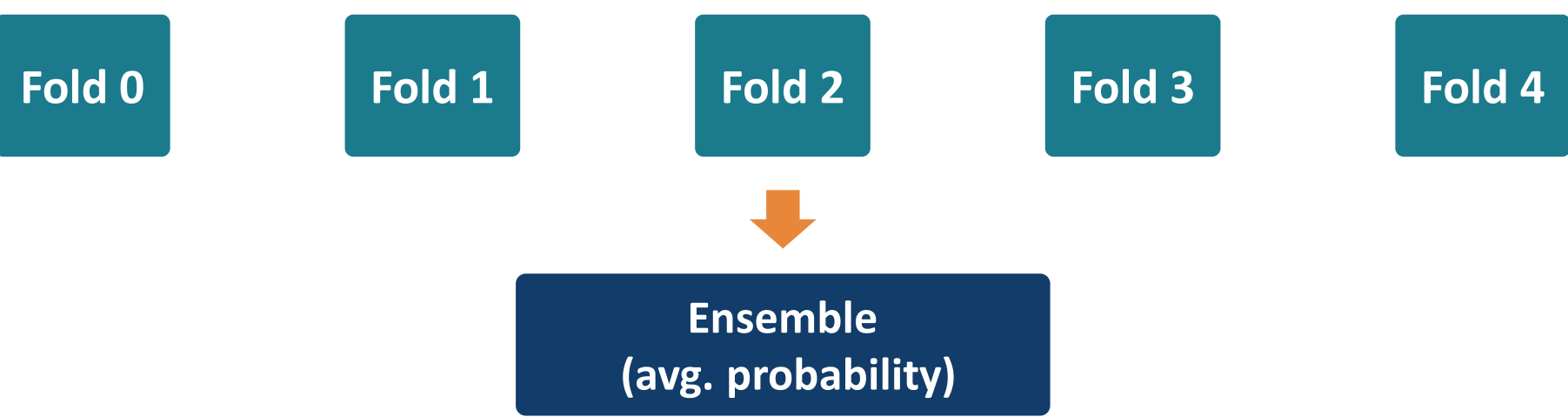


Training & Validation Strategy

- Two-phase transfer learning: frozen EfficientNetB3 backbone (≤ 30 epochs, $lr = 1e-4$), then fine-tune the top 25% of the backbone (≤ 35 epochs, $lr = 1e-5$; BatchNorm layers stay frozen).
- Label-smoothed cross-entropy weighted by damped, square-root-scaled class weights (Table 2).
- Patient-grouped, stratified 5-fold cross-validation: each fold trains an independent model; a held-out set of 329 images is never used for training or fold selection.
- Final prediction = average of the five fold models' softmax probabilities.

ACK	BCC	MEL	NEV	SCC	SEK
0.72	0.67	2.7	1.25	1.41	1.28

Table 2. Damped per-class weights used during training.



Results

73.9%

Ensemble accuracy

0.727

F1-score (weighted)

0.624

F1-score (macro)

The ensemble outperforms every individual fold model (67.2%–74.8% accuracy), confirming that averaging five patient-grouped folds yields a more reliable estimate than any single split.

Category	Precision	Recall	F1	N
ACK	0.82	0.8	0.81	104
BCC	0.72	0.86	0.78	121
MEL	0.5	0.43	0.46	7
NEV	0.84	0.77	0.81	35
SCC	0.31	0.18	0.23	28
SEK	0.7	0.62	0.66	34

Table 3. Per-class precision, recall, and F1-score of the ensemble on the held-out test set.

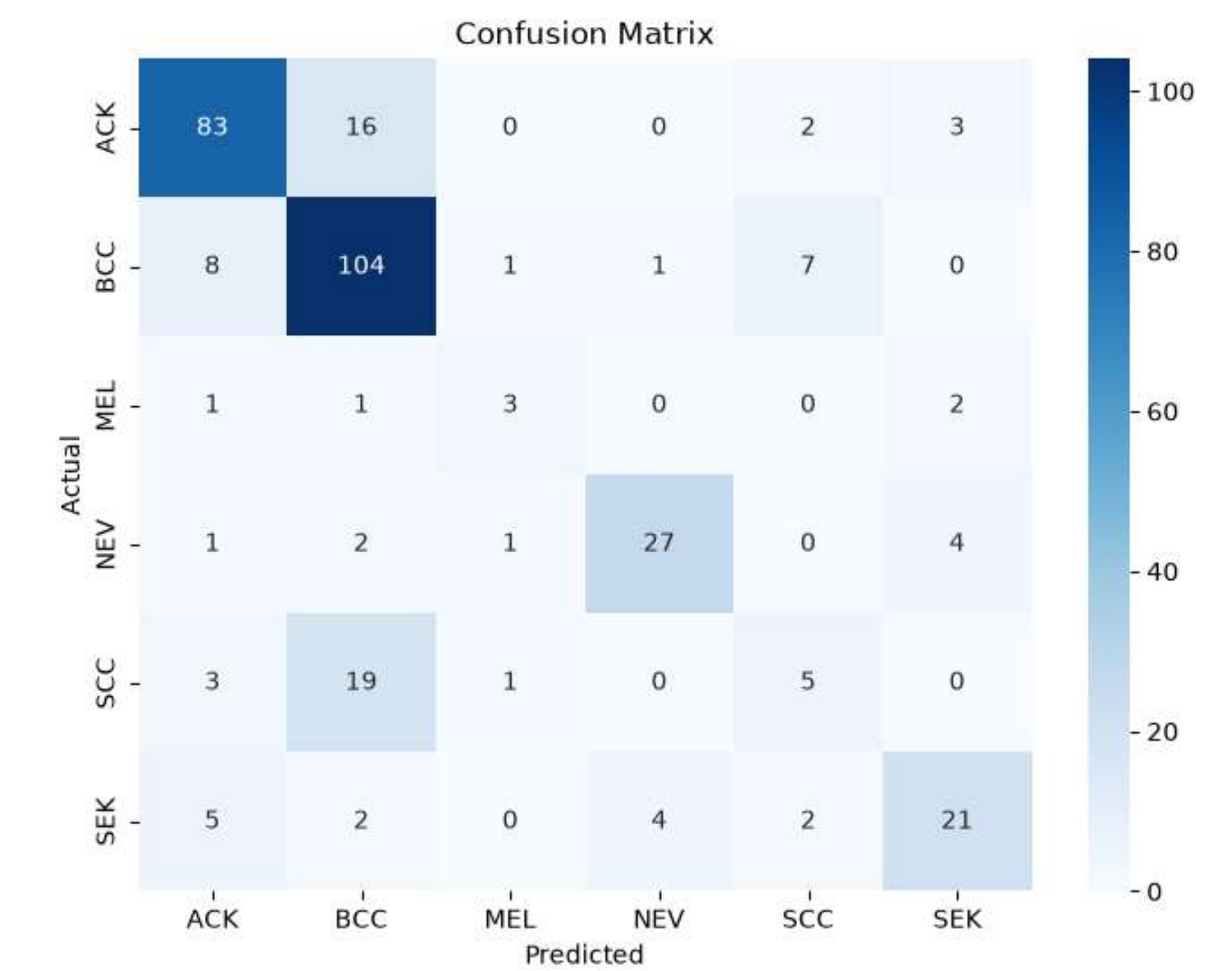


Fig. 5. Confusion matrix of the ensemble on the held-out test set.

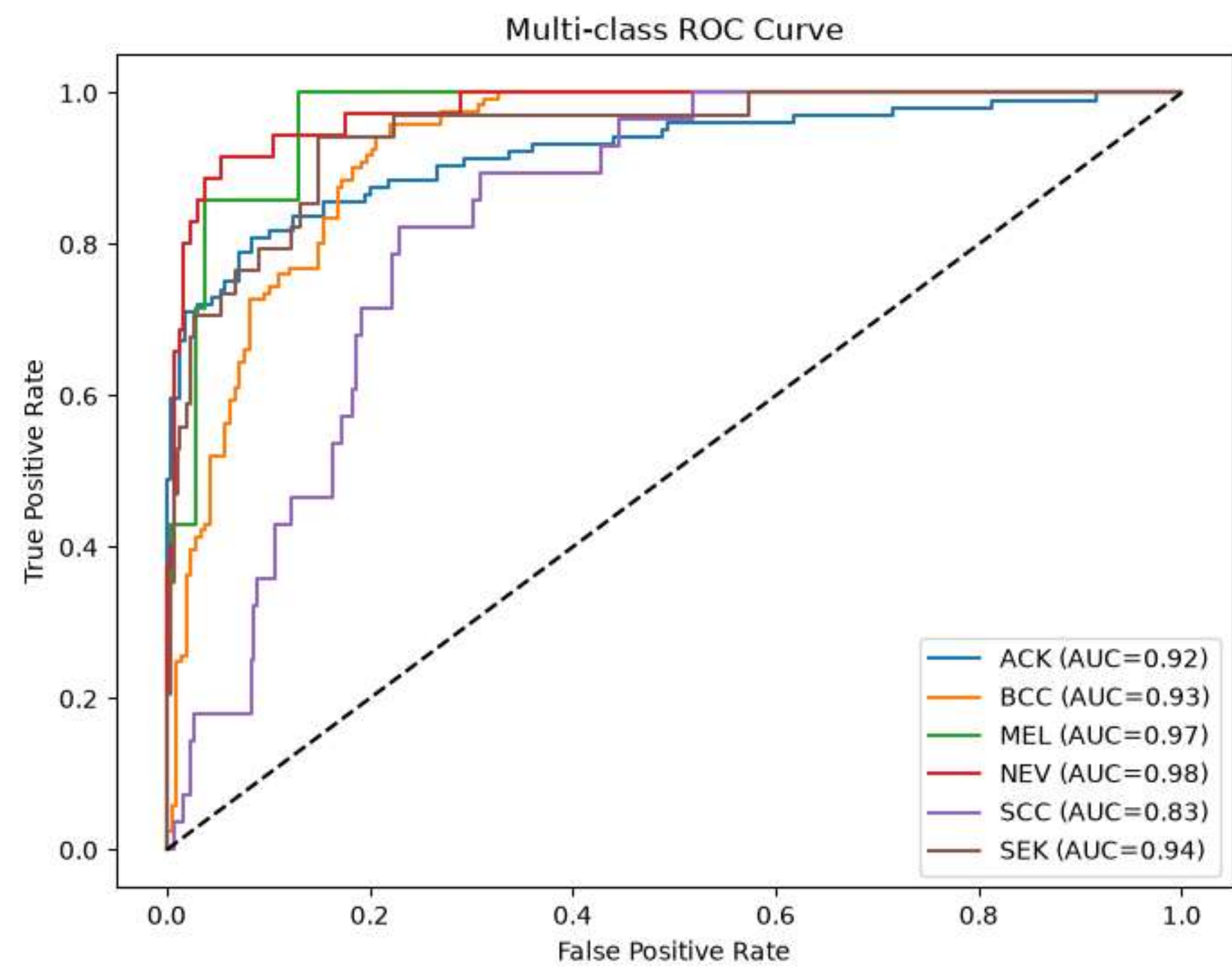


Fig. 6. Per-class ROC curves of the ensemble on the held-out test set.

Discussion & Conclusion

- Strong performance on ACK, BCC, NEV (F1 0.78–0.81); SCC remains the main limitation (F1 0.23), driven by genuine visual overlap with ACK/BCC and a limited number of training examples — not a loss-function deficiency.
- The gap between weighted F1 (0.727) and macro F1 (0.624) quantifies this class-imbalance effect.
- Limitations: single backbone architecture; no controlled ablation isolating the contribution of color constancy; very small MEL test set (7 images).
- Future work: targeted SCC data collection, backbone comparison, and an ablation of the color-constancy step.

Key takeaway: patient-level validation combined with 5-fold ensembling turns a noisy single-split estimate into a reliable ~74% accuracy benchmark for smartphone-based skin cancer screening.